Discrete Structures DSCR211

Chapter 1: Formal Logic

Introduction to Logic

Mathematics can be broadly classified into two categories –

- Continuous Mathematics
- Discrete Mathematics

Continuous Mathematics is based upon continuous number line or the real numbers. It is characterized by the fact that between any two numbers, there are almost always an infinite set of numbers. For example, a function in continuous mathematics can be plotted in a smooth curve without breaks.

Discrete Mathematics, on the other hand, involves distinct values; i.e. between any two points, there are a countable number of points. For example, if we have a finite set of objects, the function can be defined as a list of ordered pairs having these objects, and can be presented as a complete list of those pairs.

Introduction to Logic

- *Discrete Mathematics* is a branch of mathematics involving discrete elements that uses algebra and arithmetic. It is increasingly being applied in the practical fields of mathematics and computer science. It is a very good tool for improving reasoning and problem-solving capabilities.

Introduction to Logic

- Goals of Discrete Mathematics
  - *Mathematical Reasoning* – Students must understand mathematical reasoning in order to read, comprehend and construct mathematical arguments. It includes the discussion of mathematical logic which serves as a foundation for the subsequent discussion of method of proof.
  - *Combinatorial Analysis* – An important problem – solving skill is the ability to count or enumerate objects; it includes the discussion of basic techniques of counting.
  - *Discrete Structures* – a course in discrete mathematics should teach students how to work with discrete structures, which are the abstract mathematical structures used to represent discrete objects and relationships between these objects. These discrete structure include set, permutations, relations, graphs, trees, and finite – state machines.

Introduction to Logic

- *Algorithmic Thinking* – certain classes of problems are solved by the specification of an algorithm, Aftermath, an algorithm has been described a computer program can be constructed implementing it.
- *Application and Modeling* – Discrete mathematics has applications to almost every conceivable area of study. There are many applications to computer science and data networking .

Introduction to Logic

- *What is a logic?*

it is technically defined as “the science or study of how to evaluate arguments and reasoning.”

- *Logical reasoning*

it is used in mathematics to prove theorems.

- *Mathematical Logic (symbolic logic)*

it is a branch of mathematics with close connections to computer science.

Four (4) Divisions

1. Set Theory
2. Model Theory
3. Recursion Theory
4. Proof Theory

Mathematical Logic

- Definition: Methods of reasoning, provides rules and techniques to determine whether an argument is valid
- Theorem: a statement that can be shown to be true (under certain conditions)
  - Example: If x is an even integer, then $x + 1$ is an odd integer
    - This statement is true under the condition that x is an integer is true.

Mathematical Logic

- A **statement**, or a **proposition**, is a declarative sentence that is either true or false, but not both
- Uppercase letters denote propositions
  - Examples:
    - P: 2 is an even number (true)
    - Q: 7 is an even number (false)
    - R: A is a vowel (true)
  - The following are not propositions:
    - P: My cat is beautiful
    - Q: My house is big

Mathematical Logic

- Truth value
  - One of the values "*truth" (T)* or "*falsity" (F)* assigned to a statement

Examples:

1. Manila is the capital of the Philippines.
2. What day it is?
3. Help me, please.
4. He is handsome.

Logical Connectives

- **Proportional Variable**

  it is a variable which used to represent a statement.

- **Logical connectives**

  are used to combine simple statement which are referred as compound statement.

- **Compound statement**

  it is a statement composed of two or more simple statements connected by logical connectives. (and, or, if then, not, if and only if, exclusive – or)

- **Simple (atomic) statement**

  a statement which is not compound.

Logical Connectives

Six (6) main logical connectives

1. Conjunction
2. Disjunction
3. Negation
4. Conditional
5. Bi-conditional
6. Exclusive - or

Logical Connectives

- *Conjunction*
  - Let *p* and *q* be statements. The **conjunction** of *p* and *q*, written *p* ^ *q* , is the statement formed by joining statements *p* and *q* using the word "and"
    - The statement *p* ^ *q* is true if both p and q are true; otherwise

p ^ *q* is false.

- Truth Table for Conjunction:

Logical Connectives

- *Disjunction*
  - Let p and q be statements. The **disjunction** of p and q, written $p \vee q$, is the statement formed by joining statements p and q using the word "or"
    - The statement $p \vee q$ is true if at least one of the statements p and q is true; otherwise $p \vee q$ is false
    - The symbol $\vee$ is read "or"
    - Truth Table for Disjunction:

Logical Connectives

- *Negation*
  - The negation of the statement *p* is denoted by ~ *p* where ~ is the symbol for not.
    - If p is true
    - The symbol ~ is read as not. , ~ is false.
    - Truth Table for Disjunction:

Logical Connectives

- *Conditional or Implication*
  - Let p and q be statements. The statement "if p then q" is called an **implication or condition**.
  - The conditional statement $p \rightarrow q$ is false only when p is true and q is false.
  - The implication "if p then q" is written $p \rightarrow q$
  - p is called the hypothesis, q is called the conclusion
- Truth Table for Implication:

Logical Connectives

- *Bi–conditional (Bi-implication)*
  - Let *p* and *q* be statements. The statement "*p* if and only if *q*" is called the **bi-implication or bi-conditional** of *p* and *q*
  - If *p* and *q* are true or both false, then p $\leftrightarrow$ q is true; if *p* and *q* have opposite truth values, then p $\leftrightarrow$ q is false.
  - The bi-conditional "*p* if and only if *q*" is written $p \leftrightarrow q$
  - "*p* if and only if *q*"
  - Truth Table for the Bi-conditional:

Logical Connectives

- *Exclusive – or*
  - The exclusive – or of the statement *p* and *q* is the compound statement *p* exclusive – or *q*.
  - Symbolic, *p* xor *q*, where xor is the symbol for exclusive or.
  - If p and *q* are true or both false, then *p* xor *q* is false; if *p* and *q* have opposite truth values, then *p* xor *q* is true.

Logical Connectives

- Precedence of logical connectives is:
  - $\sim$ highest
  - $\wedge$ second highest
  - v third highest
  - $\rightarrow$ fourth highest
  - $\leftrightarrow$ fifth highest

Truth Tables

- Construct the truth table for each of the following statements:
  1. $\sim p \vee \sim q$
  2. $p \rightarrow (p \leftrightarrow \sim q)$
  3. $(\sim p \wedge q) \vee (p \wedge \sim q)$
  4. $(p \rightarrow q) \vee (q \wedge \sim r)$

Compound Statement

- Three (3) important classes of compound statement:
  1. *Tautology* is a compound statement that is true for all possible combinations of the truth values of the propositional variables also called logical true.
  2. *Contradiction* is a compound statement that is false for all possible combinations of the truth values of its propositional variables also called logically false or absurdity.
  3. *Contingency* is a compound statement that can be either true or false depending on the truth values of the propositional variables are neither a tautology nor a contradiction.

Compound Statement

- Example: Write the truth table for each of the following compound statements and determine whether the compound statement is tautology, contradiction or contingency.
  1. $(\sim p \wedge q) \rightarrow q$
  2. $(p \rightarrow q) \wedge (p \rightarrow \sim q)$
  3. $(\sim p \vee q)$ xor $(p \rightarrow q)$

Logical Equivalence

- *Logically Implies*
  - A statement formula p is said to *logically imply* a statement formula q if the statement formula $p \rightarrow q$ is a tautology. If p logically implies q, then symbolically we write $p \rightarrow q$.
- *Logically Equivalent*
  - A statement formula p is said to be *logically equivalent* to a statement formula q if the statement formula

    $p \leftrightarrow q$ is a tautology. If p is logically equivalent to q, then symbolically we write $p \Leftrightarrow q$.

Fundamental Logical Equivalences

Idempotence[a]

$(P \vee P) \Leftrightarrow P$

$(P \wedge P) \Leftrightarrow P$

Associativity

$[(P \vee Q) \vee R] \Leftrightarrow [P \vee (Q \vee R)]$

$[(P \wedge Q) \wedge R] \Leftrightarrow [P \wedge (Q \wedge R)]$

Commutativity

$(P \vee Q) \Leftrightarrow (Q \vee P)$

$(P \wedge Q) \Leftrightarrow (Q \wedge P)$

Distributivity ($\wedge$ over $\vee$)

$[P \wedge (Q \vee R)] \Leftrightarrow [(P \wedge Q) \vee (P \wedge R)]$

$[(P \vee Q) \wedge R] \Leftrightarrow [(P \wedge R) \vee (Q \wedge R)]$

Law of the Excluded Middle

$[P \vee (\neg P)] \Leftrightarrow \mathbf{T}$

Law of Double Negation (Involution)

$\neg(\neg P) \Leftrightarrow P$

Law of Simplification

$[(P \wedge Q) \rightarrow P] \Leftrightarrow \mathbf{T}$

$[(P \wedge Q) \rightarrow Q] \Leftrightarrow \mathbf{T}$

Domination

$(P \vee \mathbf{T}) \Leftrightarrow \mathbf{T}$

$(P \wedge \mathbf{F}) \Leftrightarrow \mathbf{F}$

Identity

$(P \vee \mathbf{F}) \Leftrightarrow P$

$(P \wedge \mathbf{T}) \Leftrightarrow P$

De Morgan's Laws

$[\neg(P \vee Q)] \Leftrightarrow [(\neg P) \wedge (\neg Q)]$

$[\neg(P \wedge Q)] \Leftrightarrow [(\neg P) \vee (\neg Q)]$

Distributivity ($\vee$ over $\wedge$)

$[P \vee (Q \wedge R)] \Leftrightarrow [(P \vee Q) \wedge (P \vee R)]$

$[(P \wedge Q) \vee R] \Leftrightarrow [(P \vee R) \wedge (Q \vee R)]$

Law of Contradiction

$[P \wedge (\neg P)] \Leftrightarrow \mathbf{F}$

Law of Addition

$[P \rightarrow (P \vee Q)] \Leftrightarrow \mathbf{T}$

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[a] An idempotent is an algebraic element for which $x^2 = x$.

Variation of Conditional Statement

- Implication
  - Let p: Today is Sunday and q: I will wash the car.
  - *Conditional :* $p \rightarrow q$:

    If today is Sunday, then I will wash the car
  - The ***converse*** of this implication is written $q \rightarrow p$

    If I wash the car, then today is Sunday
  - The ***inverse*** of this implication is written $\sim p \rightarrow \sim q$

    If today is not Sunday, then I will not wash the car
  - The ***contrapositive*** of this implication is written $\sim q \rightarrow \sim p$

    If I do not wash the car, then today is not Sunday